Practice Quiz: Lecture 7 – Part 1

Topics: UTP/STP Cabling, Shielding Designations, AWG Gauge vs. Attenuation, Wiring Standards (TIA/EIA 568A/B), Crossover vs. Straight-Through Cables, and Connectors

Student Name: _____ **Student ID:** _____ **Score:** _____ / 20

1. Among Cat3, Cat4, Cat5e, and Cat6, which twisted-pair category supports the highest transmission data rate and best transmission distance performance?

( ) A. Cat3

( ) B. Cat4

( ) C. Cat5e

( ) D. Cat6

2. True or False: A standard Unshielded Twisted-Pair (UTP) network cable contains exactly 4 individual copper wires in total.

( ) True

( ) False

3. Under the standard ISO/IEC 11801 naming scheme (XX/YTP), which twisted-pair cable features an overall braided screen combined with an overall foil screen surrounding unshielded pairs?

( ) A. U/UTP

( ) B. F/UTP

( ) C. SF/UTP

( ) D. S/FTP

4. A network technician needs to connect two enterprise routers back-to-back via their Fast Ethernet ports without an intermediary switch. Which cable configuration must be used?

( ) A. Straight-through patch cable

( ) B. Crossover cable

( ) C. Rollover (console) cable

( ) D. Fiber patch cable

5. Regarding conductor gauge specifications (AWG) in copper cabling, which statement is physically correct?

( ) A. A smaller AWG number indicates a smaller conductor diameter and higher attenuation.

( ) B. A smaller AWG number indicates a thicker conductor diameter and lower attenuation.

( ) C. Cables of identical AWG exhibit identical attenuation regardless of operating signal frequency.

( ) D. AWG measures frequency modulation bandwidth rather than physical wire gauge.

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6. What is the standard 8-pin modular connector used for terminating twisted-pair Ethernet cables?

Answer: _____

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7. True or False: Increasing the number of twists per unit length (twist density) degrades cable quality and increases signal corruption.

( ) True

( ) False

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8. Which engineering trade-off is most characteristic of choosing Shielded Twisted-Pair (STP) over Unshielded Twisted-Pair (UTP) in industrial environments?

( ) A. STP provides superior immunity against EMI and crosstalk, but is bulkier, stiffer, and less flexible to route.

( ) B. STP completely eliminates signal attenuation over unlimited distances, but requires optical transceivers.

( ) C. STP eliminates the necessity for twisting copper wires, but incurs higher manufacturing labor costs.

( ) D. STP operates exclusively with legacy proprietary mainframe connectors.

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9. Which commercial wiring termination pinout standard is most predominantly adopted and installed across local network infrastructures in Thailand?

( ) A. TIA/EIA-568A

( ) B. TIA/EIA-568B

( ) C. ISO 27001:2022

( ) D. IEEE 802.11ax

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10. What is the primary physical objective of twisting copper conductors into balanced pairs inside an Ethernet cable?

( ) A. To increase the total mechanical tensile strength of the cable assembly.

( ) B. To cancel electromagnetic interference (EMI) from external sources and minimize mutual crosstalk between adjacent pairs.

( ) C. To allow alternating current voltage to flow in a single direction.

( ) D. To prevent individual conductors from expanding due to thermal dissipation.

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11. What is the maximum certified horizontal channel length specified by TIA/EIA standards for standard twisted-pair Ethernet runs (e.g., Cat5e/Cat6)?

( ) A. 50 meters

( ) B. 100 meters

( ) C. 185 meters

( ) D. 500 meters

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12. In the TIA/EIA-568A and TIA/EIA-568B wiring standards, which two color-coded wire pairs are transposed (swapped) with each other?

Answer: Pair _____ (Orange) and Pair _____ (Green)

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13. A Category 6 cable link exhibits an attenuation rate of 20 dB per 100 meters at 250 MHz. What is the signal power loss in decibels across a shorter run of 35 meters?

Answer: _____ dB

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14. Decode the shielding structure of an S/FTP cable according to the ISO/IEC notation:

Overall cable shielding type: _____

Individual pair shielding type: _____

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15. When creating a straight-through patch cable, how must the cable ends be terminated?

( ) A. One end terminated with TIA/EIA-568A and the opposite end with TIA/EIA-568B.

( ) B. Both ends terminated identically with the same standard (either both 568A or both 568B).

( ) C. Pins 1 and 2 connected to pins 7 and 8 at the opposing end.

( ) D. Only the receive pins connected while leaving transmit pins open.

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16. What is the primary physical difference between an RJ-11 connector and an RJ-45 connector?

( ) A. RJ-11 uses optical fiber; RJ-45 uses twisted copper.

( ) B. RJ-11 is a 6-position connector used for analog telephone lines, whereas RJ-45 is an 8-position/8-contact connector used for Ethernet.

( ) C. RJ-11 has built-in shielding foil; RJ-45 is unshielded.

( ) D. RJ-11 operates at 10 Gbps; RJ-45 operates at 10 Mbps.

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17. Compare 24 AWG wire with 22 AWG wire: Which wire has a larger cross-sectional area, and which wire exhibits lower electrical resistance per unit distance?

Larger diameter: _____ AWG

Lower resistance/attenuation: _____ AWG

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18. What is the definition of Near-End Crosstalk (NEXT) in twisted-pair testing?

( ) A. Noise introduced by ambient lightning strikes measured at the destination receiver.

( ) B. The unwanted signal coupling from a transmitting pair into an adjacent receiving pair, measured at the same end of the cable link where the transmitter is located.

( ) C. The total signal attenuation measured after traversing the complete length of the link.

( ) D. Thermal noise produced by resistance within the patch panel termination blocks.

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19. What Ethernet interface capability allows two modern network devices to communicate using either a straight-through or crossover cable interchangeably?

Answer: _____

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20. What is the maximum rated operational frequency bandwidth of Category 5e versus Category 6 cabling?

Cat5e Bandwidth: _____ MHz

Cat6 Bandwidth: _____ MHz

Answer Key & Detailed Explanations (Lecture 7 – Part 1)

Question 1: D [Cat6][cite: 9]

Among the listed legacy and modern categories (Cat3, Cat4, Cat5e, Cat6), Cat6 delivers the highest certified frequency bandwidth (250 MHz) and data rates (up to 10 Gbps over shorter runs and 1 Gbps up to 100 m) with significantly reduced crosstalk[cite: 9].

Question 2: False[cite: 9]

A standard UTP Ethernet cable consists of **4 pairs** of twisted wires, which equals a total of **8 individual insulated copper wires** (conductors)[cite: 9].

Question 3: C [SF/UTP][cite: 9]

Under ISO/IEC 11801, the code format is *Outer Shield / Pair Shield* TP[cite: 9]. "SF" denotes both a braided screen (S) and a foil screen (F) around the entire cable, while "/UTP" denotes unshielded individual twisted pairs[cite: 9].

Question 4: B [Crossover cable][cite: 10]

Devices of the same functional layer transmitting and receiving on the identical pin pairs (such as Router-to-Router, Switch-to-Switch, or PC-to-PC) require a crossover cable to cross the TX lines of one device to the RX lines of the other[cite: 10].

Question 5: B [Smaller gauge, smaller attenuation][cite: 10]

In American Wire Gauge (AWG), gauge size is inversely related to physical diameter[cite: 10]. A smaller gauge number corresponds to a thicker wire with lower resistance, resulting in smaller signal attenuation[cite: 10].

Question 6: RJ-45[cite: 10]

Registered Jack-45 (RJ-45) is the standardized 8-pin, 8-contact modular physical connector used for unshielded and shielded twisted-pair computer networking[cite: 10].

Question 7: False[cite: 11]

A higher twist rate (more twists per foot or inch) improves mutual cancellation of radiated electromagnetic noise and differential crosstalk, resulting in higher cable performance and signal integrity[cite: 11].

Question 8: A [STP offers superior noise and crosstalk protection, but is bulkier and less flexible][cite: 11]

STP provides metallic shielding that blocks severe electromagnetic noise and industrial motor interference, but the added foil/braid and drain wire make the cable significantly thicker, heavier, stiffer, and more expensive to install[cite: 11].

Question 9: B [TIA/EIA-568B][cite: 11]

While both 568A and 568B are electrically identical when matched on both cable ends, TIA/EIA-568B is the de facto commercial standard most widely adopted throughout corporate and institutional premises in Thailand and North America[cite: 11].

Question 10: B [To reduce electromagnetic interference (EMI) and crosstalk][cite: 12]

By twisting the two balanced conductors together, both wires receive nearly equal electromagnetic interference from external fields[cite: 12]. Differential receivers then subtract the common noise while reconstructing the desired differential voltage[cite: 12].

Question 11: B [100 meters]

The maximum certified length for a standard Ethernet horizontal twisted-pair channel is 100 meters (typically structured as 90 meters of solid core horizontal cable plus up to 10 meters of stranded patch cord).

Question 12: Pair 2 (Orange) and Pair 3 (Green)

TIA/EIA-568A assigns Green to Pins 1 & 2 and Orange to Pins 3 & 6. TIA/EIA-568B swaps them, assigning Orange to Pins 1 & 2 and Green to Pins 3 & 6. (Pairs 1 Blue and 4 Brown remain on identical pins).

Question 13: 7.0 dB

$$\text{Attenuation} = 20\ \text{dB} \times \left(\frac{35\ \text{m}}{100\ \text{m}} \right) = 20 \times 0.35 = 7.0\ \text{dB}$$

Question 14: Braid screen (S) and Foil shield (F)

In S/FTP: 'S' indicates an overall outer braided copper screen; 'FTP' indicates that each individual pair is wrapped in its own protective foil shield (Foiled Twisted Pair).

Question 15: B [Both ends terminated identically]

A straight-through patch cable requires both ends to follow the exact same wiring standard (either 568A at both ends, or 568B at both ends) so that Pin 1 connects to Pin 1, Pin 2 to Pin 2, and so on.

Question 16: B

RJ-11 is a 6P4C/6P2C connector intended for single- or dual-line telephone systems. RJ-45 is an 8P8C connector intended for 4-pair structured Ethernet data networks.

Question 17: 22 AWG (Larger diameter); 22 AWG (Lower resistance)

Because AWG is inverse, 22 AWG has a larger diameter (approx. 0.64 mm) than 24 AWG (approx. 0.51 mm). A thicker cross-sectional area yields lower DC resistance and less attenuation over distance.

Question 18: B

NEXT measures the signal energy coupled from the transmit pair into the receive pair at the near end (the transmitter's end) of the link. It is one of the most critical parameters in cable certification.

Question 19: Auto-MDIX (Automatic Medium-Dependent Interface Crossover)

Auto-MDIX allows network PHY chips to automatically detect whether a straight-through or crossover cable is connected and configure their internal transmit and receive circuits accordingly.

Question 20: 100 MHz (Cat5e); 250 MHz (Cat6)

Category 5e is certified to test limits of 100 MHz, whereas Category 6 is certified up to 250 MHz (with more stringent crosstalk and system noise requirements).
